IAN PSHEA-SMITH, Ph.D., MPH

MEDICAL GEOGRAPHER & GLOBAL HEALTH EPIDEMIOLOGIST

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EDUCATION

Ph.D., Medical Geography and Global Health — May 2026

University of Florida, Gainesville, Florida

Awards: Verena Lighthouse Scholar, Evan Coe Medical Geography Awards (Travel & Excellence), Distinguished Teaching Award, two oral presentation awards & one poster presentation award

Dissertation: Modeling Vector Ecologies Using Frameworks from Geography, Community Ecology, and Public Health

MPH, Global Health Epidemiology — May 2023

University of Michigan, Ann Arbor, Michigan

Awards: Latin American and Caribbean Studies Tinker Fellow, Global Public Health Research Award Recipient, Outstanding Commitment to Public Health Practice

Projects: Mosquito landing behavior in northern Ecuador, Infection prevention in Grenada's Homes for the Aging, Key informant perspectives on long-COVID in San Antonio, Texas

BSc, International Health | Minors in Biology & Spanish — December 2020

Bethel University, Mishawaka, Indiana

Awards: Scholar Athlete in Track & Field, Outstanding Major in International Health

Projects: Narrative review of zoonotic malaria in South East Asia, Segregation distortion in *Drosophila melanogaster*

RESEARCH EXPERIENCE

Postdoctoral Scholar — June 2026 - Present

University of California, Berkeley, Department of Integrative Biology, Berkeley, California

- Led research on risk mapping of flavivirus spillover dynamics across central and eastern Africa
- Mentored graduate at Makrere University and at UC Berkeley in biostatics and spatiotemporal modelling
- Led grant-writing efforts and lab meetings under the direction of the PI

Graduate Research & Teaching Assistant — September 2023 - May 2026

University of Florida Department of Geography, Gainesville, Florida

- Conducted research relating to mosquito ecology and zoonotic disease using machine learning and traditional regression approaches
- Constructed and submitted proposals, manuscripts and conference presentations
- Led coursework on medical geography and served as a grader for interdisciplinary courses

Research Assistant II

April, 2023 - December, 2025

University of Michigan Department of Epidemiology, Ann Arbor, Michigan

- Conducted data analysis (R, SAS) for multiple studies on mosquito behavior patterns in northern Ecuador
- Applied modeling techniques and epidemiological theory
- Presented findings at research meetings and drafted results for publication

Global Health Intern

May 2023 - August 2023

Eversight Vision, Ann Arbor, Michigan

- Conducted a meta analysis of global attitudes towards corneal donation
- Analyzed data using Microsoft Excel and R
- Drafted results for publication and presentation, coordinating authorship and submission

Research Assistant I

September 2021 - April 2022

University of Michigan Department of Health Behavior, Ann Arbor, Michigan

- Processed data relating to a parent-focused behavioral intervention on childhood obesity
- Coded & processed qualitative data using SPSS & Dedoose
- Participated in IRB & HIPAA compliance and trainings

Science Laboratory Assistant

January 2020 - May 2021

Bethel University, Mishawaka, Indiana

- Conducted molecular genetics research on segregation distortion in *Drosophila melanogaster*
- Performed PCR, gel electrophoresis, plasmid transformation and shotgun cloning
- Assisted in routine laboratory management including inventory, culture preparation and flu husbandry

CLINICAL EXPERIENCE

Clinical Informatics Specialist

December 2020 - September 2021

Oak Street Health LaSalle Park, South Bend, Indiana

- Documented patient's medical history, medications, medical management & diagnoses
- Performed efficient & precise medical coding compliant with the Centre for Medicare's guidelines
- Lead daily team meetings presenting patients to provider and clinical team-members

Medical Registrar & COVID-19 Testing Attendant

December 2020 - May 2021

Beacon Health System, Granger & South Bend, Indiana

- Swabbed patients and performed RT-PCR assays on samples to assess for COVID-19
- Registered patients in EMRs & charted information
- Trained and supervised employees on proper procedures and EMR usage

Medical Scribe

April 2018 - May 2019

Scribe America, Elkhart General Hospital, Elkhart, Indiana

- Documented provider visits for emergency medicine physicians
- Incorporated knowledge of pharmaceuticals, medical terminology and pathophysiology
- Followed HIPPA guidelines and navigated complex power dynamics and interpersonal relationships

PUBLIC HEALTH EXPERIENCE

Epidemiology & Entomology Intern Summer 2022

Research Center for Infectious and Vector Diseases (CIREV), Quito, Ecuador

- Collected mosquitoes for identification and molecular analysis using HLC & HDN methods
- Analyzed data from mosquito collections using R and SAS
- Crafted protocols and drafted manuscripts for publication
- Assisted in mosquito husbandry associated with a sterile insect release programme

Environmental & Vector Technician Summer 2019

Elkhart County Environmental Health Department, Goshen, Indiana

- Assessed distribution of mosquitoes & ticks, alongside vector identification and control
- Created guide for pesticide use and contraindications
- Performed weekly surface-water quality analyses

Intercultural Intern Fall 2019

Devon Oasis, Little India, Chicago, Illinois

- Provided social support and basic health/nutritional education to incoming refugee families
- Developed course materials for TESOL and health education
- Supervised homework tutoring sessions and assisted with college applications

TEACHING EXPERIENCE

University of Florida Instructor of Record

GEO 3452 | Introduction to Medical Geography

- Four semesters, 30-75 students per semester
- Managed and led online course, graded all course material, designed course content

University of Florida Teaching Assistant

IDS 2935 | The Next Pandemic

- Three semesters, 150-200 students per semester
- Graded weekly assignments, quizzes, exams and presentations

University of Michigan Graduate Student Instructor

SOC 302 | Health & Society

- Two semesters, 75 students per semester
- Lectured on supplementary information, led discussion sections and graded essays

Bethel University Teaching Assistant & Grader

BIOL 220 | Clinical Microbiology with Laboratory

- One semester, 2 20-student sections
- Prepared media and cultures for procedures, supervised laboratory sessions and graded assignments

BIOL 115 | Human Biology with Laboratory

- One semester, 3 25-student sections
- Prepared materials and equipment for procedures, supervised laboratory sessions and graded assignments

SPAN 161 - 162 | Elementary Spanish I & 2

- Four semesters, 4 20-student sections per semester
- Graded weekly quizzes, semester exams and short writing assignments

INVITED TALKS, TRAININGS & WORKSHOPS

1. **Modelling vector ecologies using frameworks from community ecology, geographic sciences and public health** | January 2026, Invited talk, Annual Meeting of the Institute for Infectious Disease Translational Research, University of South Carolina; 2026
2. **Genomic Surveillance and Epidemiological Models for Outbreak Response Workshop** | May 2025, Emerging Pathogens Institute, University of Florida, Gainesville, Florida
3. **Global Health Epidemiology Panel** | EPID 664 - Field Methods in Global Health Epidemiology, January 2023, Department of Epidemiology, University of Michigan, Ann Arbor, Michigan

SCIENTIFIC PRESENTATIONS

1. **Pshea-Smith IA**, Kotorashvili A, Kotaria N, Golubiana G, Kirkitadze G, Chunashvili T, Shubashishvili A, Di Paola N, Kugelman J, Hulseberg C, Walker B, Musich T, Cleary NG, Metrailer MC, Reinbold-Wasson D, Denlinger DS, Blackburn JK, von Fricken ME. An ecological analysis of ticks in the country of Georgia using ten years of surveillance data.
   Poster presentation
   Conference of Research Workers in Animal Diseases
   Chicago, Illinois; 2026
2. **Pshea-Smith IA**, Andritsch S, Kotorashvili A, Kotaria N, Golubiana G, Kirkitadze G, Chunashvili T, Shubashishvili A, Di Paola N, Kugelman J, Hulseberg C, Walker B, Musich T, Reinbold-Wasson D, Blackburn JK, von Fricken ME. Species distribution models for *Ixodes ricinus* and *Dermacentor reticulatus* in the Republic of Georgia.
   Poster presentation
   Annual meeting of the American Society of Tropical Medicine and Hygiene
   Toronto, Canada; 2025

3. **Pshea-Smith IA**, Okech B, Boncy J, Existe A, Metrailer M, Matulis GA, Blackburn JK, von Fricken ME. Mosquito biodiversity in a pathogen endemic setting in Haiti.
   Poster presentation
   Emerging Pathogens Institute Research Day
   Gainesville, Florida; 2025

4. **Pshea-Smith IA**, Okech B, Sutherland I, Dunford J, So J, Koehler J, Boncy J, Matulis G, Ponciano JM, Blackburn JK, von Fricken ME. Mosquitoes & Haiti: Modelling mosquito distribution and examining mosquito diversity under current and future climate scenarios.
   Oral presentation
   Annual meeting of the American Association of Geographers
   Detroit, Michigan; 2025

5. **Pshea-Smith IA**, Werner E, Dunn M, Bashshur R. Geographic trends of keratitis in Florida and its association with social vulnerability.
   Oral presentation
   Florida Society of Geographers Annual Meeting
   Gainesville, Florida; 2025

6. **Pshea-Smith IA**, Okech B, Sutherland I, Dunford J, So J, Koehler J, Boncy J, Matulis G, Ponciano JM, Blackburn JK, von Fricken ME. Estimating spatial distributions of *Aedes aegypti*, *Aedes albopictus* and *Culex quinquefasciatus* in Haiti using machine learning.
   Poster presentation
   Annual meeting of the American Society of Tropical Medicine and Hygiene
   New Orleans, Louisiana; 2024

7. **Pshea-Smith IA**, Okech B, Sutherland I, Dunford J, So J, Koehler J, Boncy J, Matulis G, Ponciano JM, Blackburn JK, von Fricken ME. Modelling and predicting counts and environmental correlates of *Aedes aegypti*, *Aedes albopictus* and *Culex quinquefasciatus* in Haiti.
   Oral presentation
   International Medical Geography Symposium
   Atlanta, Georgia; 2024

8. **Pshea-Smith IA**, Okech B, Sutherland I, Dunford J, So J, Blackburn JK, von Fricken ME. Spatiotemporal Patterns of Three Mosquito Species in Haiti, 2018-2019.
   Poster presentation
   Emerging Pathogens Institute Research Day 2024
   Gainesville, Florida; 2024

9. **Pshea-Smith IA**, Ponce P, Cevallos V, Carrazco A, Morrison AC, Coloma J, Eisenberg JNS. Determinants of *Aedes aegypti* & *Culex* Species Landing Rates in Northern Esmeraldas Province, Ecuador: A Human Landing Catch Study, June 2021 - July 2022.
   Poster presentation
   Annual meeting of the American Society of Tropical Medicine and Hygiene
   Chicago, Illinois; 2023

SCIENTIFIC PUBLICATIONS

Accepted & Published

1. **Pshea-Smith IA**, Ponce P, Cevallos V, Carrazco A, Morrison AC, Coloma J, Eisenberg JNS. Landing patterns of *Aedes aegypti* & *Culex quinquefasciatus* in northern Esmeraldas province, Ecuador: Human Landing Catch Study June 2021 - July 2022.
   Accepted, Journal of the American Mosquito Control Association; in proof
   Preprint: https://doi.org/10.1101/2025.03.12.25323448

2. Bashshur R, Werner E, **Pshea-Smith IA**, Dunn M. Prevalence and geographic distribution of keratitis in the United States and associated factors.
   Accepted, Next Research, April 2026

3. **Pshea-Smith IA**, Okech BA, Boncy J, Existe A, So J, Hamilton T, Blanton J, Matulis GA, Metrailer MC, Cleary NG, Dunford JC, Koehler JW, Blackburn JK, von Fricken ME. Vegetation, temperature, and human population density influence mosquito biodiversity in Haiti. *Parasites & Vectors*. 2026 Oct 9. https://doi.org/10.1186/s13071-026-07724-3

4. Matulis GA, Smith HP, Katich RS, Kour N, Delp KL, Douglas CE, **Pshea-Smith IA**, Lilak AA, Okech B, Existe A, Sutherland I. Short report: surveillance of *hrp2/hrp3* deletions in *Plasmodium falciparum* within Haiti, 2021-2023, using RDTs. *Malaria Journal*. 2026 Aug 28;25:313. https://doi.org/10.1186/s12936-026-06108-3

5. **Pshea-Smith IA**, Bissainte D, Ash KD. Characterising climate change, colonisation, conflict & corruption as a vulnerabilising assemblage in the context of infectious disease in Haiti. *Geoforum*. 2026 Aug;174:104683. https://doi.org/10.1016/j.geoforum.2026.104683

6. **Pshea-Smith IA**, Okech B, So J, Boncy J, Sutherland I, Hamilton T, Dunford J, Blanton J, Existe A, Rudolph FJ, Matulis GA, Ponciano JM, Koehler JW, Blackburn JK, von Fricken ME. Modeling and predicting counts and environmental correlates of mosquito distributions in Haiti using surveillance data from the Ouest Department. *Am J Trop Med Hyg*. 2026 May 12:tpmd250324. https://doi.org/10.4269/ajtmh.25-0324

7. Matulis GA, Smith HP, Hall G, Katich RS, Delp KL, Douglas CE, Williams J, Hirsch S, Kumar R, **Pshea-Smith IA**, Lilak AA, Okech BA, Ricks K, Stefan CP, Existe A, Kugelman JR, Sutherland I, Dunford J, Boncy J, von Fricken ME, Koehler JW. Characterization of arbovirus infections in patients within Haiti by screening discarded malaria rapid diagnostic test cassettes. *PLOS Neglected Tropical Diseases*. 2026 Mar 20;20(3):e0014089. https://doi.org/10.1371/journal.pntd.0014089

8. Motes J, Wilson B, **Pshea-Smith IA**, Matulis GA, Bird JT, So J, Boncy J, Sutherland IW, Hamilton T, Dunford J, Scime S, Tarigopula N, Minard-Smith A, Existe A, Koehler JW, Okech BA, von Fricken ME, Blanton J. Molecular detection of insecticide resistance markers in *Aedes aegypti* (Diptera: Culicidae) populations from Haiti. *Journal of Medical Entomology*. 2026 Feb 6;63(1):tjaf195. https://doi.org/10.1093/jme/tjaf195

9. **Pshea-Smith IA**, Bashshur M, Itani W, Mofrad ZN, Stout E, Bashshur R. Perceptions on Corneal and Eye Donation: Implications for Corneal Donor Programs. Eye Banking and Corneal Transplantation. 2024 Sep 1;3(3):e0029. https://doi.org/10.1097/ebct.0000000000000029

Under Review

1. **Pshea-Smith IA**, Okech B, Sutherland I, Dunford J, So J, Boncy J, Existe A, Punch B, Matulis GA, Koehler JW, Blackburn JK, von Fricken ME. The increasing suitability of *Aedes aegypti* in Haiti under climate change is an issue of global health equity.
   Under review, Journal of Climate Change & Health

2. **Pshea-Smith IA**, Kotorashvili A, Kotaria N, Golubiani G, Kirkitadze G, Chunashvili T, Shubashishvili A, Di Paola N, Kugelman J, Hulseberg C, Walker B, Musich T, Cleary N, Metrailer MC, Blanton J, Lynn J, Reinbold-Wasson D, Denlinger DS, Blackburn JK, von Fricken ME. An ecological analysis of ticks in the country of Georgia.
   Under review, Ecosphere
   https://doi.org/10.2139/ssrn.6551591

3. Cleary N, Fiorenzano J, Boldbaatar B, **Pshea-Smith IA**, Altantogtokh D, Lindroth EJ, Davidson S, Poole-Smith BK, Hertz J, Amgalanbayar B, Etobayeva I, Salyer AR, Letizia AG, Ryan SJ, von Fricken ME. Modeling the distributions of *Dermacentor nuttalli*, *Ixodes persulcatus*, and *Hyalomma asiaticum* in Mongolia from 2019-2024.
   Under review, Ticks and Tick-borne Diseases
   https://doi.org/10.2139/ssrn.6327699

PROFESSIONAL REPORTS & DOCUMENTS

1. **Pshea-Smith IA***, Gardner K*, Harrison T*, Tretyakova S*. Dementia Management and Infectious Disease Prevention in Grenada's Homes for the Aging. 2023. Internal report for Grenada's Ministry of Social & Community Development, Housing and Gender Affairs. *All four authors contributed equally to this report.

2. **Pshea-Smith IA***, Vyas J*, Fry J*. Long COVID & Health Information in Bexar County. 2023. Internal report for the South-Central Advanced Health Education Center in San Antonio, Texas. *All three authors contributed equally to this report.

3. **Pshea-Smith IA***, Tretyakova S*, Grandinetti K*, Powers K*, Tanksley A*. Grenada Red Cross Volunteering Policy. 2022. Internal use document for the Grenada Red Cross Society. *All five authors contributed equally to this report.

4. **Pshea-Smith IA**. Elkhart County Pesticide Guide. 2021. Internal use document for the Elkhart County Environmental Health Department.

PROFESSIONAL MEMBERSHIPS

2025 - Present | Florida Society of Geographers
2024 - Present | American Association of Geographers
2024 - Present | Society of Vector Ecology
2021 - Present | American Society of Tropical Medicine and Hygiene

FELLOWSHIPS & AWARDS

2025 | **College of Liberal Arts and Sciences Travel Award** - University of Florida
2025 | **Doctoral Student Poster Presentation Winner** - Emerging Pathogens Research Day
2025 | **Distinguished Teaching Award as a Graduate Student** - University of Florida
2025 | **Evan Coe Award for Excellence in Medical Geography** - University of Florida
2025 | **Student Oral Presentation Winner** - Ann. Meeting of the Florida Society of Geographers
2024 | **Student Paper Presentation Winner** - International Medical Geography Symposium
2024 | **Evan Coe Medical Geography Travel Award** - University of Florida
2023 | **Verena Lighthouse Scholar** - NSF Biology Integration Institute
2023 | **Outstanding Commitment to Public Health Practice** - University of Michigan PHAST
2022 | **Global Public Health Research Award** - University of Michigan
2022 | **Latin American and Caribbean Studies Tinker Fellow** - University of Michigan
2020 | **Outstanding Major in International Health** - Bethel University
2017-2020 | **Scholar Athlete**, Track & Field - Bethel University

TECHNICAL SKILLS

- Data Analysis: R, SAS, ArcGIS Pro
- Microbiology & molecular biology: PCR, electrophoresis, asceptic & sterile technique
- Mosquito identification, husbandry for SIT and environmental control
- Survey development and deployment, semi-structured and focus group interviewing

PROFESSIONAL SKILLS

- Cultural competency & sensitivity
- Pedagogy in interdisciplinary fields, lesson plan development and discussion facilitation
- Professional Spanish proficiency; beginner French proficiency
- Grant/proposal writing and project management

INTERESTS

Academic

- Vector ecology
- One Health
- Public health practice
- Spatial analysis + visualization

Personal

- Long distance running
- Kayaking
- Science fiction & fantasy books